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TEACHERS WITHOUT BORDERS PROGRAMME

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Herzlia High	Rondebosch Boys'	St Mary's DSG Kloof	Wynberg Secondary

GRADE 9 MATHEMATICS
JUNE 2019

Time: 2 Hours **Total: 120**

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 5 pages. Please check that your paper is complete.
2. Read the questions carefully.
3. Answer all the questions.
4. Number your answers exactly as the questions are numbered.
5. You may use an approved non-programmable and non-graphical calculator, unless otherwise stated.
6. Round off your answers to one decimal digit where necessary.
7. All the necessary working details must be clearly shown. Answers only, without relevant calculations, may incur penalties.
8. It is in your own interest to write legibly and to present your work neatly.

Question 1

1.1 Write 0,000 000 000 679 543 in Scientific notation. (2)

1.2 Express $2{,}34\times10^{-4}$ as a number (2)

1.3 Simplify $(5{,}123\times10^{4})\times(2\times10^{-2})$, leave your answer in Scientific notation (3)

1.4 Simplify, leaving your answer with a positive exponent:

1.4.1. $-2a^{3}b\times6a^{4}b^{2}$ (2)

1.4.2 $(2x^{2}y^{4})^{3}$ (2)

1.4.3 $\sqrt{16a^{16}}$ (2)

1.4.4 $\left(\dfrac{2x^{3}}{x^{-2}y}\right)^{2}$ (3)

[16]

Question 2

2.1 Expand and simplify:

2.1.1 $(a-3)(a+1)$ (2)

2.1.2 $(x+3)^{2}-(2x-1)(2x+1)$ (4)

2.2 Factorise completely:

2.2.1 $3p^{2}q+15pq^{2}-12pq$ (2)

2.2.2 $75x^{3}-12x$ (3)

2.2.3 $3x(x-3)+2(3-x)$ (3)

2.2.4 $2x+8+bx+4b$ (3)

2.3 Find the value of $2x^{2}+5x-12$ if $x=-3$. (2)

2.4 If $5^{y}=k$, find 5^{y+1} in terms of k. (3)

[22]

Question 3

3.1 Simplify:

3.1.1 $\frac{2x^3y^3}{2x^4} \times \frac{4xy^3}{6y} \times \frac{3x^2}{xy^3}$ (4)

3.1.2 $\sqrt[3]{\frac{54x^6}{2x^3}} - \sqrt{\frac{8x^2y^3}{2y}}$ (4)

3.1.3 $\frac{5m}{m^2-25} \div \frac{m+3}{m^2+2m-15}$ (4)

3.2 Solve for x:

3.2.1. $3(x+1) = 2x-3$ (3)

3.2.2 $\frac{2x+1}{3} = 5 - \frac{1}{2}x$ (5)

3.2.3 $2^{x+1} = 16$ (3)

3.2.4. $\left(\frac{1}{9}\right)^{x-2} = 81$ (4)

3.3 Monica received x marks in a test. Sandra received 4 marks more than Monica.

3.3.1 Write down Sandra's marks in terms of x. (1)

3.3.2 When Monica subtracts 7 from her mark and squares the result, her answer is 1 more than Sandra's mark.

Write down an equation in x and show that it simplifies to

$$x^2 - 15x + 44 = 0$$ (4)

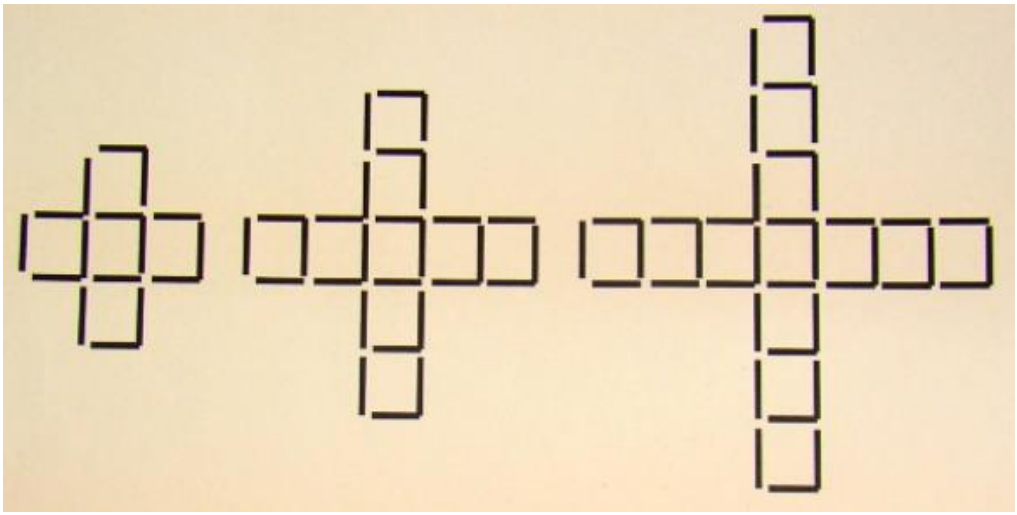
3.3.3 Solve the equation $x^2 - 15x + 44 = 0$. (3)

3.3.4 The test was marked out of 10. Write down the mark each girl received. (3)

[38]

Question 4

4.1 The following diagrams were built using sticks:



Picture 1 Picture 2 Picture 3

4.1.1 Study the diagrams above. Give the correct values of (a), (b) and (c).

Picture number (n)	1	2	3	4	5	6	…	(c)
Number of sticks (S)	16	28	40	52	(a)	(b)	…	124

(3)

4.1.2 Write down an equation for the *n*th term. (3)

4.1.3 Use your formula to calculate which picture has 1228 sticks in total. (2)

4.2 Given the pattern 2 ; 8 ; 32 ; 128

4.2.1 Give the next two terms. (2)

4.2.2 Write down an equation for the nth term. (3)

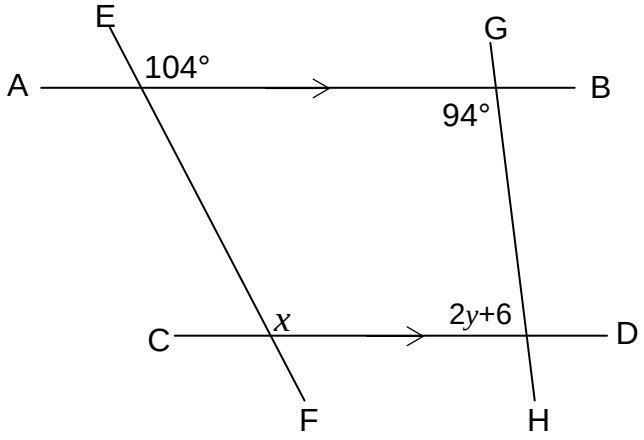
4.2.3 Find the value of the 10th term? (3)

[16]

Question 5

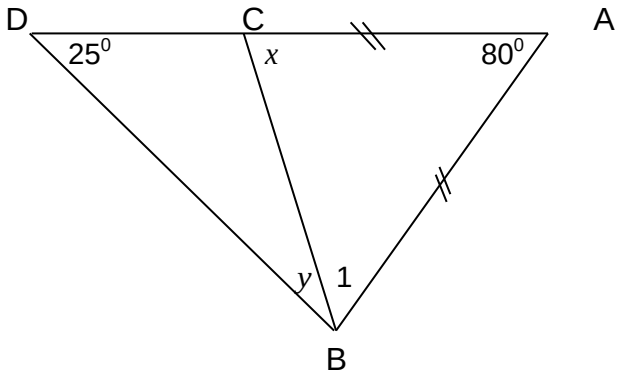
5.1 Solve for x and y in the diagrams below. Give reasons for your answers.

5.1.1



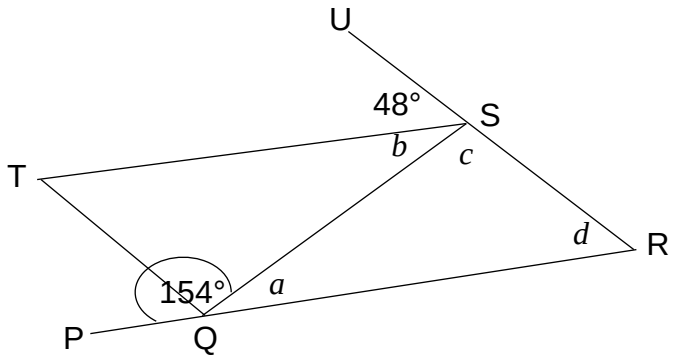
(5)

5.1.2



(8)

5.2 In the diagram below QRST is a parallelogram. Solve for the unknowns ($a - d$) in the diagram below. Give reasons for your answers



(9)

5.3 The foot of a ladder is 10 m from the wall. The length of the ladder is 2 m longer than the height that it reaches on the wall. What is the length of the ladder?
(Hint: Let x = the height) (6)

[28]